

EXCLUSIVE

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**SCIENCE
&
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- Rare Earth Elements are a series of seventeen substances that are present in the Earth's crust. Rare Earths occur plentifully in nature, but the rarity comes from the ability to isolate them chemically and make them usable in industrial applications.
- Heavy and Light Rare Earths occur naturally in several countries such as India, China, Myanmar, Japan, Australia and North Korea.
- China several times, imposed curbs on the exports of dysprosium, gadolinium, lutetium, samarium, scandium, terbium, and yttrium. The above-mentioned Rare Earth Elements are used in hard disks and car motors, nuclear reactors, medical equipment, PET scanners, powerful magnets in personal electronics and also in the fighter aircrafts.
- While Rare Earths are found in multiple countries, refining them, particularly heavy REEs like the ones described above, has been an area dominated by China. Practically, all heavy REEs used all over the world are refined in China.



- Rare Earth Elements are key ingredients of electric cars, rockets, computer chips, even vacuum cleaners. Hence, a supply choke will hurt industries.
- Tesla cars and other EVs, for instance, can't do without super-strong magnets.
- Critically, US and everyone else needs them for military applications like drones that have proved their worth in the Ukraine war.
- China has built a monopoly on rare earth processing and magnet manufacturing. Now, China is in a position to pay back United States of America for tariffs and the AI chip ban.

WHAT IS THE INDIAN GOVERNMENT'S PLAN FOR RARE EARTH ELEMENTS?



- As a policy framework for utilizing critical minerals, including Rare Earth Metals, the National Critical Mineral Mission (NCMM) has been launched, which is India's strategic initiative to secure critical mineral supply chain by increasing domestic critical minerals production and foreign supply sources.
- Critical minerals are a larger umbrella under which REEs fall - in a 2022 list of 30 critical minerals notified by the Union Government, all 17 REEs were one item.
- Under the NCMM, the government has said it will facilitate or engage in 1,200 exploration projects, award exploratory licenses to incentivize private exploration, and auction more critical mineral blocks.
- The mission also aims to streamline regulatory approvals for critical mineral exploration and extraction.

How China consolidated its hold in critical minerals such as Copper, Cobalt, Nickel, Lithium and Rare Earth Elements (REEs)?



- China has channeled \$56.9 billion through its State-backed institutions since the year 2000 to secure extraction and processing operations in 19 BRI countries - primarily in Africa, Latin America and Southeast Asia.
- Chinese firm acquired a cobalt-copper mine in the DRC. In the DRC, Chinese entities control 51% of the cobalt exports through JVs.
- China process 73% of the World's cobalt and 59% of lithium.
- China is Peru's largest copper investor investing in two important Toromocho and La Bambas copper mines. Now, Chinese firms are dominating mining sectors in Argentina (Lithium), Zambia (Copper) and Indonesia (Nickel).

81% of China's loans for critical minerals qualify as non-public debt directed not at its sovereign government, but at Special Purpose Vehicles (SPVs) and Joint Ventures (JVs) with Chinese equity stakes.



- PFAS stands for Per-and Polyfluoroalkyl Substances (PFAS).
- These are a group of chemicals used to make fluoropolymer coatings and products that resist heat, oil stains, grease, and water.
- These products are used in clothing, furniture, adhesives, food packaging, heat resistant nonstick cooking surfaces, etc.
- Many PFAS are a cause of concern because:
 - ✓ They do not breakdown in the Environment.
 - ✓ They can move through soils and contaminate drinking water sources.
 - ✓ They build up (bio-accumulate) in Fish and Wildlife.

These are called “forever chemicals” because they take thousands of years to biodegrade.

How are people exposed to PFAS? How do these chemicals affect people's health?



- PFAS persist in the environment and exposure in people can occur by consuming contaminated water or food.
- Exposure may also happen by using products that contain PFAS.
- Human health effects from exposure to low levels are uncertain. However, when exposure is at high levels, they may affect growth and development - reproduction, thyroid function, immune system and injury to the liver.



- Last year, the Food Safety and Standards Authority of India (FSSAI), which is under the Ministry of Health and Family Welfare released a logo for these Vegan Foods.
- Whenever we are purchasing the foods, we can look for the logo issued by FSSAI.
- Vegan foods does not include milk and milk related products as well as egg and egg related products.
- Even honey and honey related products are also not part of these Vegan Foods.





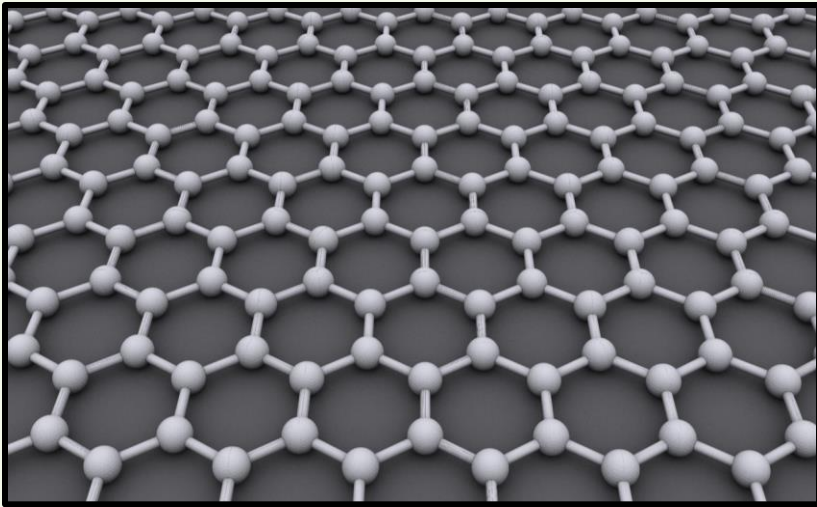
- Most of us are fond of fast foods, we can also call them junk food.
- There are two serious health concerns with regard to fast foods:
 - ✓ First one is with regard to the way they are prepared - most of them are prepared through deep frying. The oil is used again and again, because of which the Total Polar Compounds (TPCs) in the oil exceed the limit of 25%. They are harmful, once they exceed 25%.
 - ✓ Second one is Trans-fats. Most of the fast foods have industrially produced Trans-fats. They raise the bad cholesterol and harmful for heart.





- Across the World, the temperatures are increasing due to global warming and one of the prominent reasons is the increase in the **Green House Gases** like **Carbon Dioxide**, **Methane**, **Nitrous Oxide**, among others.
- Because of **cattle rearing**, lot of **Methane** is **released** into the **environment**. Beef is one of the **animal products** **consumed** extensively **across** the **World**.
- Now, to reduce global warming, alternatives are being thought of. **Stem cells from animals** are **taken**, and they are **cultured** and **grown** in the **laboratories** so as to give texture of animal meat, and this is known as cultured meat.





- Around 70 years ago, it was the beginning of plastics age or polymers age. Around 30-40 years ago, Semiconductors era started.
- As per experts, the future era is going to be the Graphene era.
- Graphene is considered to be the wonder material; it is thinnest on the Earth - just one atom thick with hexagonal lattice structure of one Carbon atom thick. A sheet of one square meter of graphene weighs 0.77 milligrams. It is pure carbon.
- It is 200 times stronger than steel and its electrical conductivity is much higher than Copper. It has the highest heat conductivity.
- Because of all these characteristics, it is going to see wider applications in the future, as per experts.
- It is used in electronics, energy storage, sensors, coatings, composites, biomedical devices and many other applications. Its high surface area and biocompatibility make it an attractive material for drug delivery and tissue engineering applications.

Why do we need Supercomputers? What are Airawat, Param-Siddhi, Pratyush and Mihir?

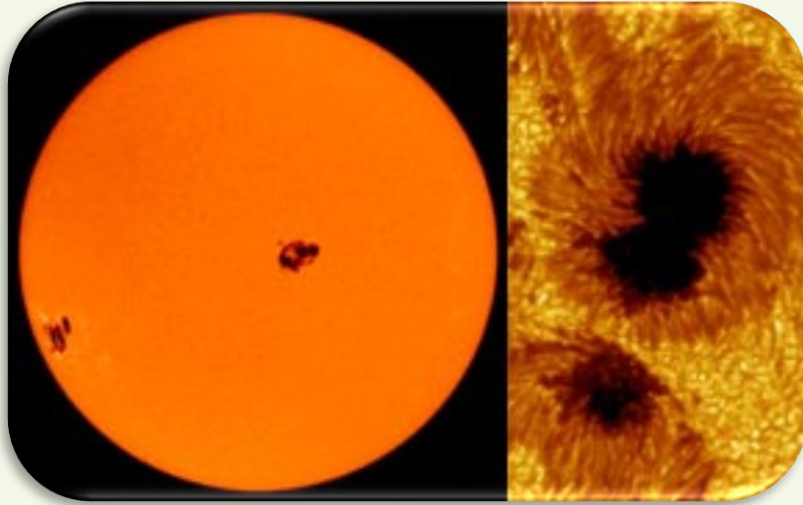


- There are several complexities in weather forecasting and cell biology. If we are developing new medicines, we should look at different types of proteins, which are long chains of amino acids. Similarly Artificial Intelligence based applications require huge capability. Hence, Supercomputers are required.
- The definition of a Supercomputer is not constant, and it is in a flux. Around 20-25 years ago, the computing power was very less, but now it increased tremendously. Now, the measure of the Supercomputers is in exaflops i.e., 10^{18} FLOPs and petaflops i.e., 10^{15} FLOPs.
- TOP500 is the designation given by an organization listing Top 500 Supercomputers in the World and this list is updated once in six months. Frontier of United States of America is the fastest Supercomputer.
- For India, Airawat is at 75th rank followed by Param-Siddhi. Airawat and Param-Siddhi are used for AI applications whereas, Pratyush and Mihir are used for weather forecasting.



- Tidal locking is the phenomenon by which a body has the same rotational period as its orbital period around a partner. For example, Moon is tidally locked to the Earth because it rotates in exactly the same time as it takes to orbit the Earth. In other words, it takes precisely as long for the Moon to spin on its axis as it does to complete its monthly orbit around Earth.
- All the solar system's large Moons are tidally locked with their planets.
- Because of tidal locking, we always see the same side of the Moon, which we call the near side. The other hemisphere, which we never see is called the far side of the Moon.

The lunar hemisphere facing the Earth will always face the Earth and the hemisphere facing away will always face away.



- The Solar magnetic cycle, that works in the deep interior of the Sun creates regions, that raise to the surface and appear like dark spots. These are the 'Sunspots'.
- Solar flares are highly energetic phenomenon that happen inside the Sunspots. In a Solar flare, the energy stored in the Sun's magnetic structure is converted into light and heat energy. This causes the emission of high energy X-ray radiation and highly accelerated charged particles to leave the Sun's surface.
- The flares are seen as bright areas on the Sun, and they can last from minutes to hours. We see the Solar Flare by the photons or light it releases.

Solar wind is a stream of charged particles released from the upper atmosphere of the Sun, called the Corona.

WHAT DO YOU UNDERSTAND BY CORONAL MASS EJECTIONS (CMEs)? WHAT IS THE IMPACT OF SOLAR FLARES AND CMEs?



- Sometimes, Solar flares also cause hot plasma to be ejected from the Sun. This is called Coronal Mass Ejection (CME). CMEs can harbour energies exceeding that of a billion atomic bombs.
- The solar flares / CMEs can affect Earth bound objects and life on Earth. They can affect the electronics within satellites and affect astronauts.
- Very powerful Earth directed Coronal Mass Ejections (CMEs) can cause failure of power grids and affect oil pipelines and deep-sea cables.

The energy radiation and high energy particles can also cause Spectacular Aurorae in the high latitudes in the polar countries.



- RISC-V is an open-source Instruction Set Architecture (ISA) originally written by computer scientists at the University of California, Berkeley.
- ISA refers to the set of instructions that a computer processor can understand and execute. It acts as an interface between the hardware and the software.
- In open-source, code is public and anyone is free to take it, modify it, share it, suggest improvements, or add new features.



There were expectations few years ago about challenges from RISC-V to Intel (which has grip on desktops and data centers) and ARM (which monopolizes the market for tablet and smartphone chips).

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